Continuous Tensor Representation for Hyperspectral Anomaly Detection

Yiming Zeng, Xi-Le Zhao, *Member, IEEE*, Teng-Yu Ji, Wei-Hao Wu, Degang Wang, and Lina Zhuang, *Member, IEEE*

***Abstract*—Hyperspectral anomaly detection (HAD) is an important task in remote sensing for identifying pixels with anomalous spectral signatures that deviate from their local background. Recently, low-rank and sparse representation-based methods have garnered significant attention in HAD, which typically employ low-rank representation to characterize the background and sparse representation to capture anomalies. Since the background and anomalies usually exhibit complex characteristics beyond the low-rankness and sparsity, low-rank and sparse representation-based methods typically do not perform satisfactorily for complex scenarios. To address the challenge, we propose an unsupervised HAD method from a continuous perspective, which organically integrates continuous background representation and deep anomaly representation (CBAR). Specifically, the CBAR model leverages the continuous low-rank tensor function to encapsulate both the low-rankness and smoothness of the background and the deep neural network to capture the complex geometric structure of anomalies. Moreover, to mitigate the overfitting of the background and anomalies to the observed hyperspectral image (HSI), we introduce two terms as overfit shield by exploiting the prior knowledge of the background and anomalies. To solve the CBAR model, we develop an efficient alternating minimization algorithm. Extensive experiments on benchmark datasets [including Airport, Urban, Beach, and Hyperspectral Digital Imagery Collection Experiment (HYDICE)] demonstrate that the proposed CBAR outperforms the state-of-the-art anomaly detection methods both qualitatively and quantitatively. For reproducibility, we will release our source code at: https://github.com/Weihao-Wu/CBAR**

***Index Terms*—Deep neural network, hyperspectral anomaly detection (HAD), hyperspectral images (HSIs), low-rank tensor representation.**

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I. Introduction

HYPERSPECTRAL images (HSIs) are 3-D data comprising extensive spatial and spectral information, which has led to various applications, including change detection [1], [2], classification [3], [4], and anomaly detection [5], [6], [7]. Among them, hyperspectral anomaly detection (HAD) aims to distinguish anomalous targets from the observed HSI without prior knowledge of the spectral signal of anomalies, which is an important application in military and civilian fields. However, the HAD problem is ill-posed, and its solutions are usually unstable or nonunique. To solve the HAD problem, it is necessary to introduce suitable regularizers. Based on regularizers used in HAD, current HAD methods are primarily categorized into model-based methods [8], [9], [10], [11], [12], [13], [14], [15] and deep learning-based methods [16], [17], [18], [19], [20], [21], [22], [23].

A. Model-Based Methods

Model-based methods primarily employ various regularizers to separately characterize the background and anomalies. Based on the regularizers used to characterize the background, model-based methods can be categorized into sparse representation-based methods, collaborative representation-based methods, and low-rank representation-based methods. Sparse representation-based methods and collaborative representation-based methods factorize the background pixel into the product of a dictionary and a coefficient vector [24]. Sparse representation-based methods emphasize sparsity by imposing the l_1 norm constraint on the coefficient vector [8], [9], [10], [11]. In contrast, collaborative representation-based methods employ the l_2 norm constraint on the coefficient vector to leverage the collective contribution of all dictionary atoms [25], [26], [27]. However, these methods detect anomalies on a pixel-by-pixel basis, which neglect the correlation of all the pixels in the HSI [28].

In recent years, low-rank representation methods have gained increasing attention owing to their ability to capture the global information of HSIs [29], [30], [31]. These methods primarily employ low-rank representation to characterize the background and sparse representation to capture anomalies. For example, based on robust principal component analysis

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[32], Sun et al. [33] decomposed the observed HSI into a background matrix and an anomaly matrix, employing the matrix nuclear norm to characterize the background and the l_1 norm to capture anomalies. Zhang et al. [29] employed the Mahalanobis distance between the pixels and the low-rank background to detect anomalies in the HSI. Xu et al. [28] utilized a background dictionary and a low-rank coefficient matrix to characterize the background while employing the $\ell_{2,1}$ norm to capture sparse anomalies. Niu and Wang [30] introduced a learned dictionary to characterize the low-rank background matrix in order to improve its robustness. Cheng and Wang [34] leveraged the matrix nuclear norm, graph regularization, and total variation (TV) regularization to capture the background, which can preserve the local geometrical structure and spatial relationships. However, the aforementioned methods convert a 3-D HSI into a 2-D matrix, which inevitably destroy the intrinsic structural information of HSI.

To fully preserve the 3-D structural information of HSI, many low-rank tensor representation-based methods [35], [36], [37], [38] have been proposed. Based on the t-product [39], [40], Sun et al. [36] decomposed the 3-D HSI data into a background tensor and an anomaly tensor, where the tensor nuclear norm and the sparse norm are exploited to characterize the background and anomalies, respectively. Wang et al. [12] captured the background by using a tensorial dictionary and a coefficient tensor, where the weighted tensor nuclear norm is constrained on coefficients. Qin et al. [41] proposed a unified nonconvex surrogate function to approximate the tensor tubal rank, capturing the low rankness of the background. Based on the tensor decomposition, Feng et al. [42] adopted a tensor ring decomposition and TV regularization to exploit the low rankness and the piecewise smoothness of the background. Xiao et al. [43] decomposed the observed HSI into three components: background, anomalies, and noise, where the background is approximated using the multiscale entanglement renormalization ansatz tensor network, anomalies are characterized by group sparsity constraint, and the noise is constrained by the Frobenius norm. Yu et al. [44] segmented the HSI into superpixels and applied low-rank and sparse representation to center pixels and weighted low-rank and collaborative representation to edge pixels. Kong et al. [45] proposed a tensor double nuclear norm-based tensor approximation model, which can approximate the low rankness and the nonlocal self-similarity of background at the same time. However, due to the limited representation capability of handcrafted regularizers, model-based methods are unsatisfactory in complex scenarios.

B. Deep Learning-Based Methods

Deep learning-based methods primarily utilize a deep neural network to reconstruct the background and employ the residual difference between the observed HSI and the reconstructed background to assess anomalies [46], [47], [48], [49], [50], [51]. For example, Wang et al. [46] leveraged fully convolutional autoencoder (AE) with skip connections to reconstruct the background, with anomalies appearing as reconstruction errors. Xie et al. [49] proposed a novel AE and adversarial-learning based semisupervised background estimation model, which consists of an encoder, a decoder, and a discriminator to thoroughly capture the background. Fan et al. [52] embedded a graph constraint into the latent space of the AE, which maintains the geometric structure of the background. Wang et al. [53] proposed a pixel-shuffle downsampling blind-spot reconstruction network for HAD, which adopts a blind-spot architecture with the receptive field to avoid incorrectly reconstructing anomalies during the background reconstruction. Wu et al. [54] introduced the concept of masked AEs into fully convolutional neural network (CNN) and further developed a sparse-inspired masked CNN, which achieves an accurate reconstruction of the background and highlights anomalies in the residual map. Lian et al. [55] proposed a novel gated transformer network for HAD, which leverages content similarity to guide the reconstruction of both the background and anomalies. Fu et al. [56] introduced a multiscale mamba reconstruction network for HAD, designed to improve background representation while minimizing the reconstruction of anomalies. Wang et al. [57] proposed a Fourier conditional mask-based mixed attention method for HAD to suppress high-frequency anomalies while preserving low-frequency background information. Wu et al. [58] proposed an asymmetric integration of KAN and CNN AE network to recover the background. Wang et al. [59] proposed a vision transformer-based foundation model scalable to over one billion parameters, which has demonstrated promising performance in anomaly detection. However, deep learning-based methods often lack interpretability and fail to incorporate the inherent prior knowledge of the background and anomalies, which limits their effectiveness in HAD.

C. Motivation

Model-based methods are limited in representation capability to capture the intricate characteristics of the background and anomalies, and deep learning-based methods often lack interpretability and fail to effectively integrate inherent prior knowledge of HSI. To address these limitations, we propose an unsupervised HAD model from a continuous perspective, which seamlessly integrates continuous background representation and deep anomaly representation (CBAR). The CBAR model fully incorporates prior knowledge of both background and anomalies and offers powerful representation capacity. Specifically, the continuous background representation leverages an unsupervised low-rank tensor function [60] to effectively characterize the inherent low-rankness and smoothness of the background. Simultaneously, the deep anomaly representation utilizes an unsupervised deep neural network to capture the complex geometric structure of anomalies. The continuous background and deep anomaly representations complement each other, offering superior anomaly detection performance.

The contributions of this article are summarized as follows.

1) To faithfully characterize the background and anomalies, we propose CBAR model from a continuous perspective for the first time, which organically integrates CBAR.

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The continuous background representation characterizes both the low rankness and smoothness of the background, while the deep anomaly representation captures the complex geometric structure of anomalies.

2) To mitigate the overfitting issue, we introduce two terms as overfit-shield by exploiting the prior knowledge of the background and anomalies. Extensive experiments on benchmark datasets demonstrate that the proposed CBAR model outperforms the state-of-the-art methods both qualitatively and quantitatively.

The remainder of this article is organized as follows. Section II introduces the necessary notations. Section III describes the proposed CBAR model. Experimental results and discussions are presented in Section IV. Finally, conclusions are drawn in Section V.

II. Notations

In this article, scalars, vectors, matrices, and tensors are denoted by lowercase letters (e.g., x), bold lowercase letters (e.g., $\mathbf{x}$), bold uppercase letters (e.g., $\mathbf{X}$), and calligraphic uppercase letters (e.g., $\mathcal{X}$), respectively. For a third-order tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, we use $\mathcal{X}(i,:,:)$, $\mathcal{X}(:,j,:)$, and $\mathcal{X}(:,:,k)$ to denote the ith horizontal slice, jth lateral slice, and kth frontal slice, respectively. The (i,j,k)th element of the tensor $\mathcal{X}$ is represented by $\mathcal{X}(i,j,k)$. The ℓ_1-norm of $\mathcal{X}$ is defined as $\|\mathcal{X}\|_1 = \sum_{i,j,k} |\mathcal{X}(i,j,k)|$, and the Frobenius norm of $\mathcal{X}$ is defined as $\|\mathcal{X}\|_{\mathrm{F}} = \left(\sum_{i,j,k} |\mathcal{X}(i,j,k)|^2\right)^{1/2}$. The unfolding operator of a tensor $\mathcal{X}$ along the ith mode $(i = 1,2,3)$ is defined as $\mathrm{unfold}_i(\cdot) : \mathbb{R}^{n_1 \times n_2 \times n_3} \to \mathbb{R}^{n_i \times \prod_{j \neq i} n_j}$, which returns the unfolded matrix along mode i. This matrix is denoted by $\mathbf{X}^{(i)} = \mathrm{unfold}_i(\mathcal{X})$. The operator $\mathrm{fold}_i(\cdot)$ is the inverse of $\mathrm{unfold}_i(\cdot)$. The mode-i tensor-matrix product is defined as $\mathcal{X} \times_i \mathbf{A} = \mathrm{fold}_i(\mathbf{A}\mathbf{X}^{(i)})$. The tensorization operator of a tensor $\mathcal{X}$ is defined as $\mathcal{L}(\cdot) : \mathbb{R}^{n_1 n_2 n_3 \times 1} \to \mathbb{R}^{n_1 \times n_2 \times n_3}$.

III. Proposed Method

In this section, we first present a general framework of HAD. We then introduce the proposed CBAR model, offering a detailed exploration of its key components: continuous background representation, deep anomaly representation, and two terms of overfit shield. Finally, we describe the solution algorithm and present the corresponding pseudocode.

A. Framework of HAD

The task of HAD is to identify pixels with spectral characteristics that significantly deviate from their local background when the target spectrum is unavailable. Specifically, HAD aims to decompose the observed HSI $\mathcal{Y} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ into a background tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and an anomaly tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, which can be mathematically formulated as follows:

$$\mathcal{Y} = \mathcal{X} + \mathcal{A}. \tag{1}$$

However, separating the anomalies $\mathcal{A}$ from the observed HSI $\mathcal{Y}$ is an ill-posed problem, which requires appropriate regularizations for an effective solution. Traditional HAD methods primarily use low-rank and sparse regularizations to constrain the background and anomalies, respectively. However, these handcrafted regularizations are insufficient to fully capture the complex characteristics of the background and anomalies beyond the low rankness and sparsity, making them unsuitable for more complex scenarios.

B. CBAR Model

To faithfully characterize the background and anomalies, we propose the CBAR model for HAD from a continuous perspective for the first time, which integrates the continuous background representation and the deep anomaly representation organically. The continuous background representation employs an unsupervised low-rank tensor function to characterize the background, effectively capturing its low-rankness and smoothness. Specifically, the low-rank tensor function is denoted as $f_{\theta_{\mathcal{X}}}(\cdot)$, where $\theta_{\mathcal{X}}$ is the learnable parameter, and the function $f_{\theta_{\mathcal{X}}}(\cdot)$ maps a coordinate (i,j,k) to the corresponding background pixel value $\mathcal{X}(i,j,k)$. The deep anomaly representation utilizes an unsupervised neural network to capture the complex geometric structure of anomalies, denoted as $f_{\theta_{\mathcal{A}}}(\cdot)$, where $\theta_{\mathcal{A}}$ is the learnable parameter. $f_{\theta_{\mathcal{A}}}(\cdot)$ takes a random vector $\mathbf{z}$ as the input and outputs the anomaly tensor $f_{\theta_{\mathcal{A}}}(\mathbf{z})$. Mathematically, the CBAR model is formulated as

$$\begin{aligned} &\min_{\theta_{\mathcal{X}},\theta_{\mathcal{A}}} \|\mathcal{Y} - \mathcal{X} - \mathcal{A}\|_{\mathrm{F}}^2 + \gamma_1 \|\mathcal{X}\|_{\mathrm{BOS}} + \gamma_2 \|\mathcal{A}\|_{\mathrm{AOS}} \\ &\text{s.t. } \mathcal{X}(i,j,k) = f_{\theta_{\mathcal{X}}}(i,j,k),\ \mathcal{A} = f_{\theta_{\mathcal{A}}}(\mathbf{z}) \end{aligned} \tag{2}$$

where $\mathcal{Y} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, and $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ denote the observed HSI, background tensor, and anomaly tensor, respectively; the terms $\|\cdot\|_{\mathrm{BOS}}$ and $\|\cdot\|_{\mathrm{AOS}}$ are overfit shields designed to mitigate overfitting of the background and anomalies to the observed HSI, and are defined in Section III-E; and γ_1 and γ_2 are the tradeoff parameters of background overfit shield (BOS) and anomaly overfit shield (AOS), respectively. The flowchart of the CBAR model is illustrated in Fig. 1. In Sections III-C–III-E below, we provide detailed descriptions of the key components of the proposed CBAR model.

C. Continuous Background Representation

The background typically exhibits low rankness and smoothness, as illustrated in Fig. 2, both of which contribute to improving detection performance. Existing methods rely on handcrafted regularizations (e.g., low-rank tensor representations) to capture the properties of the background from a discrete perspective, which suffer from limited representation capability. As a result, discrete background representations are often inadequate in complex scenarios. To address the limitation of discrete background representation, we introduce the continuous background representation to faithfully characterize the inherent low rankness and smoothness of the background. To the best of our knowledge, this is the first attempt to characterize the background from a continuous perspective in HAD field.

We now formally introduce the continuous background representation. The continuous background representation employs a low-rank tensor function to map the coordinate of

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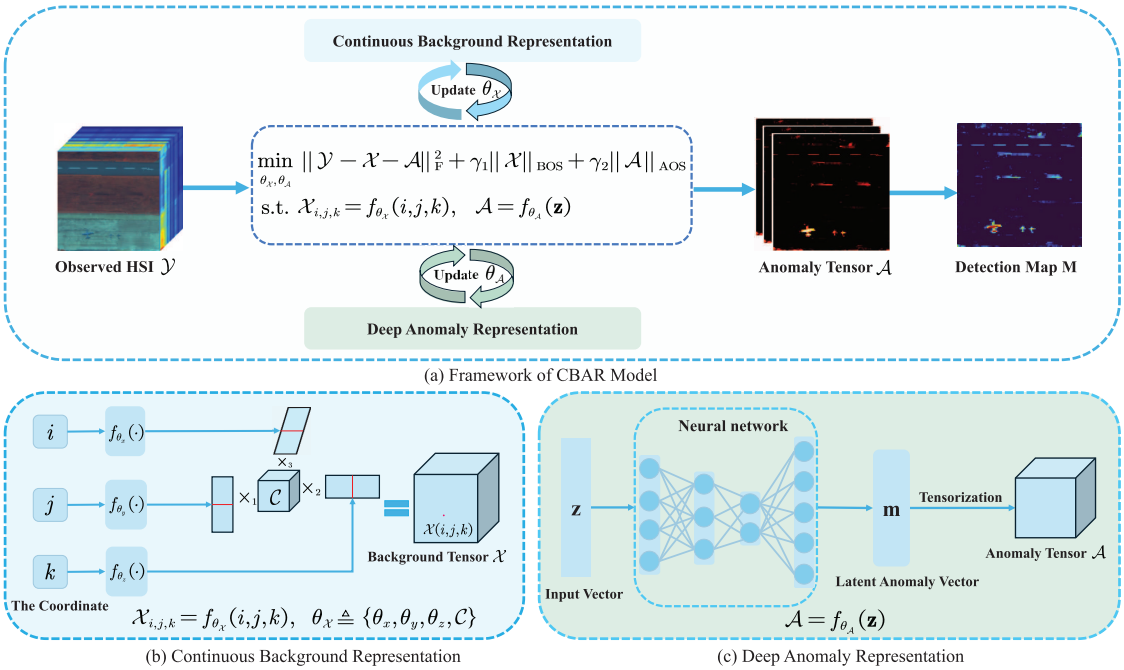


Fig. 1. Flowchart of the CBAR model. (a) Overall framework of CBAR, which decomposes the observed HSI $\mathcal{Y}$ into the background $\mathcal{X}$ and anomalies $\mathcal{A}$. (b) Continuous background representation $f_{\theta_{\mathcal{X}}}(\cdot)$, which maps the coordinate (i, j, k) to the corresponding background value $\mathcal{X}(i, j, k)$, where $\theta_{\mathcal{X}}$ includes the parameters of the three factor functions (i.e., θ_x, θ_y, and θ_z) and the parameterized core tensor $\mathcal{C}$. (c) Deep anomaly representation, which maps a random vector $\mathbf{z}$ to the anomaly tensor $\mathcal{A}$ via a deep neural network $f_{\theta_{\mathcal{A}}}(\cdot)$.

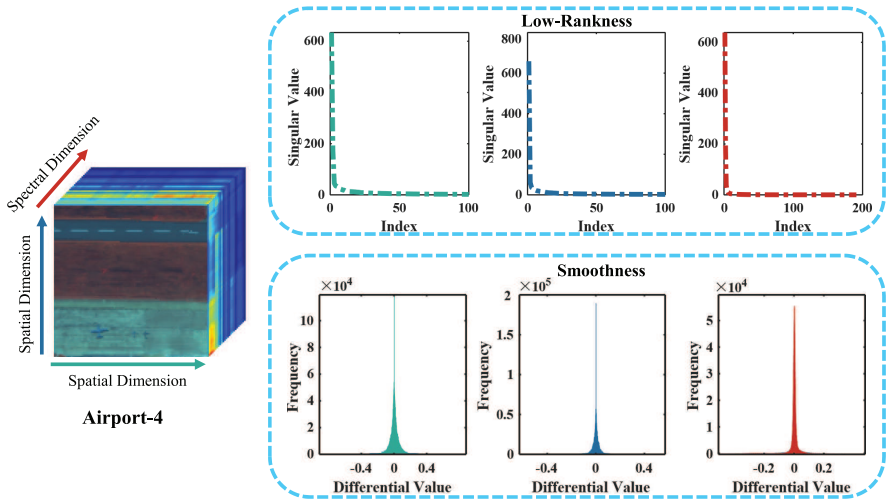


Fig. 2. Visualization of the low-rankness and smoothness of airport-4. (Top row) Singular values, illustrating the low-rank nature of the data across three dimensions. (Bottom row) Displays the frequency of differential values between adjacent pixels, highlighting the smoothness across the three dimensions.

each pixel to its corresponding background value. Specifically, the low-rank tensor function can be compactly represented by a core tensor $\mathcal{C}$ and three factor functions, i.e., $f_{\theta_x}(\cdot)$, $f_{\theta_y}(\cdot)$, and $f_{\theta_z}(\cdot)$. Mathematically, it can be formulated as follows:

$$\mathcal{X}(i,j,k)=f_{\theta_{\mathcal{X}}}(i,j,k)=\mathcal{C}\times_1 f_{\theta_x}(i)\times_2 f_{\theta_y}(j)\times_3 f_{\theta_z}(k) \tag{3}$$

where $\mathcal{C}\in\mathbb{R}^{r_1\times r_2\times r_3}$ is the core tensor, $\theta_{\mathcal{X}}\triangleq\{\mathcal{C},\theta_x,\theta_y,\theta_z\}$ denotes all the parameters, (r_1, r_2, r_3) is the predefined tensor function rank, and $f_{\theta_x}(\cdot)$, $f_{\theta_y}(\cdot)$, and $f_{\theta_z}(\cdot)$ are three factor functions. The three factor functions are bounded mappings from coordinates to vectors, defined as $f_{\theta_x}(\cdot): I_f\to\mathbb{R}^{r_1}$, $f_{\theta_y}(\cdot): J_f\to\mathbb{R}^{r_2}$, and $f_{\theta_z}(\cdot): K_f\to\mathbb{R}^{r_3}$, respectively. Here, $I_f=\{1,2,\ldots,n_1\}$, $J_f=\{1,2,\ldots,n_2\}$, and $K_f=\{1,2,\ldots,n_3\}$. Leveraging the universal approximation capability of multilayer perceptrons (MLPs) [61], we parameterize the factor functions $f_{\theta_x}(\cdot)$, $f_{\theta_y}(\cdot)$, and $f_{\theta_z}(\cdot)$ using three MLPs with learnable parameters θ_x, θ_y, and θ_z, respectively. For example, the function $f_{\theta_x}(\cdot)$ can be mathematically formulated as

$$f_{\theta_x}(i):=\mathbf{W}_d(\sigma(\mathbf{W}_{d-1}\cdots\sigma(\mathbf{W}_1 i))) \tag{4}$$

where d denotes the number of layers; $\theta_x\triangleq\{\mathbf{W}_l\}_{l=1}^{d}$ is the set of learnable weight matrices, with $\mathbf{W}_1\in\mathbb{R}^{h_1\times 1}$, $\mathbf{W}_l\in\mathbb{R}^{h_l\times h_{l-1}}$

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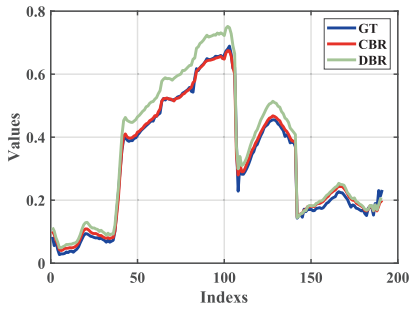


Fig. 3. Spectral vectors of background tensors obtained using different representations at spatial position (28, 88) on airport-4. GT refers to the ground truth spectral vector, CBR represents the continuous background representation, and DBR denotes the discrete background representation (discrete Tucker decomposition).

for $2 \le l \le d-1$, and $\mathbf{W}_d \in \mathbb{R}^{r_1 \times h_{d-1}}$; and $\sigma(\cdot)$ is the sine activation function. Similarly, $f_{\theta_y}(\cdot)$ and $f_{\theta_z}(\cdot)$ are defined in the same manner.

The continuous background representation $f_{\theta_{\mathcal{X}}}(\cdot)$ is compactly parameterized by a core tensor and three factor functions, which can map any coordinate (i, j, k) to the corresponding background tensor value $\mathcal{X}(i, j, k)$. This representation is inherently continuous and exhibits low rankness. Besides, when the spatial or spectral coordinate i, j, k is close, the corresponding background values $\mathcal{X}(i, j, k)$ of the continuous background representation vary smoothly. This indicates that the continuous background representation is smooth. Therefore, our continuous background representation can simultaneously characterize the low rankness and smoothness of the background, which is absent in discrete low-rank tensor representations. As shown in Fig. 3, the spectral vector of the background tensor obtained by the continuous background representation better approximates the ground truth, providing a more accurate depiction of the true underlying spectral characteristics. The flowchart of the continuous background representation is shown in Fig. 1(b).

D. Deep Anomaly Representation

Anomalies often exhibit complex geometric structures in real-world scenarios. However, handcrafted sparse representations are limited in representation capability to characterize complex anomalies. To overcome this issue, we introduce the deep anomaly representation that leverages the powerful representation capability of the deep neural network to capture the complex geometric structure of anomalies. For simplicity, we employ an MLP to capture anomalies, which consists of fully connected layers and activation functions. The selection of other architectures (e.g., CNN) is discussed in Section IV-E. The deep anomaly representation with MLP as the network architecture is formulated as

$$\mathcal{A} = f_{\theta_{\mathcal{A}}}(\mathbf{z}) = \mathcal{L}\left(\mathbf{H}_k\left(\sigma\left(\mathbf{H}_{k-1}\cdots\sigma\left(\mathbf{H}_1\mathbf{z}\right)\right)\right)\right) \tag{5}$$

where k denotes the number of layers, $\theta_{\mathcal{A}} \triangleq \{\mathbf{H}_l\}_{l=1}^{k}$ denotes the set of learnable parameters, with $\mathbf{H}_1 \in \mathbb{R}^{h_1 \times 1}$, $\mathbf{H}_k \in \mathbb{R}^{h_l \times h_{l-1}}$ for $2 \le l \le k-1$, and $\mathbf{H}_k \in \mathbb{R}^{n_1 n_2 n_3 \times h_{d-1}}$; $\sigma(\cdot)$ is the sine activation function; and $\mathcal{L}(\cdot)$ is the tensorization operator. The deep anomaly representation $f_{\theta_{\mathcal{A}}}(\cdot)$ maps a random vector $\mathbf{z} \in \mathbb{R}^{n_1+n_2+n_3}$ to the corresponding anomaly tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$.

The advantage of the deep anomaly representation lies in the powerful representation capabilities of the unsupervised deep neural network, effectively capturing the complex geometric structure of anomalies. As shown in Fig. 4, we display anomaly detection maps obtained by different representations. We observe that deep anomaly representation achieves better results in characterizing the complex geometric structure of anomalies compared to the handcrafted sparse representation. The flowchart of the deep anomaly representation is shown in Fig. 1(c).

E. Overfit Shields of Background and Anomalies

During the iterative optimization process, anomalous pixels may be reconstructed as background, while background pixels may be identified as anomalies, thereby affecting the final detection performance. To mitigate this overfitting, we introduce two terms as overfit shield by exploiting the prior knowledge of the background and anomalies: the BOS and the AOS.

The background exhibits long-range smoothness in spatial domain, which can be exploited to effectively avoid reconstructing anomalies. Therefore, the BOS is defined as

$$\|\mathcal{X}\|_{\mathrm{BOS}} = \sum_{i=1}^{n_1-1}\left\|\sum_{k=0}^{K_1}\nabla_1\mathcal{X}(i+k,:,:)\right\|_1 + \sum_{j=1}^{n_2-1}\left\|\sum_{k=0}^{K_2}\nabla_2\mathcal{X}(:,j+k,:)\right\|_1$$

where K_1 and K_2 in BOS control the group size of spatial difference group sparsity of the background along the horizontal and vertical dimensions, respectively; ∇_1 and ∇_2 represent the first-order differences operator along the horizontal and vertical dimensions, respectively. Additionally, the elements of $\nabla_1\mathcal{X}(i+k,:,:)$ or $\nabla_2\mathcal{X}(:,j+k,:)$ are set to zero if $i+k \ge n_1-1$ or $j+k \ge n_2-1$.

Anomalies exhibit group sparsity in the spatial domain, which can be exploited to avoid reconstructing background pixels. Thus, the AOS is defined as

$$\|\mathcal{A}\|_{\mathrm{AOS}} = \sum_{i=1}^{n_1}\left\|\sum_{k=0}^{K_3}\mathcal{A}(i+k,:,:)\right\|_1 + \sum_{j=1}^{n_2}\left\|\sum_{k=0}^{K_4}\mathcal{A}(:,j+k,:)\right\|_1$$

where K_3 and K_4 in AOS control the group size of group sparsity of the anomalies along the horizontal and vertical dimensions, respectively; the elements of $\mathcal{A}(i+k,:,:)$ or $\mathcal{A}(:,j+k,:)$ are zeros, if $i+k \ge n_1$ or $j+k \ge n_2$.

Since these two terms constrain the background and anomalies in their respective physically interpretable spaces, they can mitigate the overfitting. The contributions of BOS and AOS are discussed in Section IV-D.

F. Alternating Minimization-Based Solving Algorithm

To solve problem (2), we develop an alternating minimization algorithm, which decomposes the overall optimization into two simple subproblems with clear physical interpretations (the background subproblem and the anomaly subproblem). The details are as follows.

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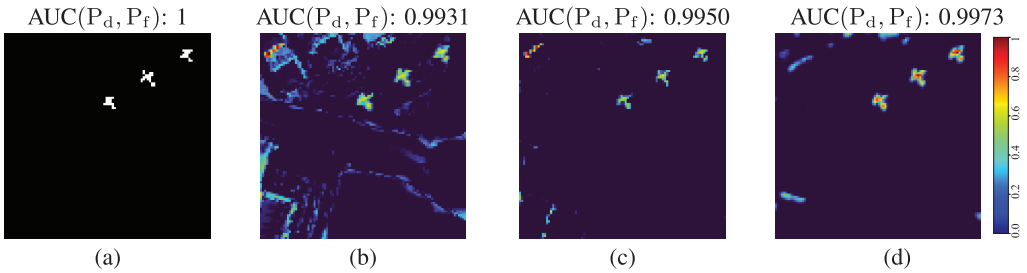


Fig. 4. Anomaly detection maps are detected by different anomaly representations with the same continuous background representation. (a) Ground truth. (b) Sparse representation: anomalies represented using handcrafted sparse representation. (c) Deep anomaly representation with MLP: anomalies detected using the deep anomaly representation with MLP as the network architecture. (d) Deep anomaly representation with CNN: anomalies detected using the deep anomaly representation with CNN as the network architecture.

1) Updating $\theta_{\mathcal{X}}$: The background tensor is characterized by the continuous background representation $f_{\theta_{\mathcal{X}}}(\cdot)$. Therefore, the background tensor can be determined by updating $\theta_{\mathcal{X}}$. At the tth iteration, the $\theta_{\mathcal{X}}$ subproblem is formulated as

$$\begin{aligned} &\min_{\theta_{\mathcal{X}}} \left\|\mathcal{Y}-\mathcal{X}-\mathcal{A}^{t}\right\|_{\mathrm{F}}^{2}+\gamma_{1}\|\mathcal{X}\|_{\mathrm{BOS}} \\ &\text{s.t. } \mathcal{X}(i,j,k)=f_{\theta_{\mathcal{X}}}(i,j,k). \end{aligned} \tag{6}$$

Since the problem (6) is highly nonlinear and nonconvex, we resort to the adaptive moment estimation (Adam) [62] to update parameter $\theta_{\mathcal{X}}$. Then, the (i,j,k)th element of background tensor can be computed as

$$\mathcal{X}^{t+1}(i,j,k)=f_{\theta_{\mathcal{X}}^{t+1}}(i,j,k). \tag{7}$$

2) Updating $\theta_{\mathcal{A}}$: The anomaly tensor is captured by the deep anomaly representation $f_{\theta_{\mathcal{A}}}(\cdot)$; thus, the anomaly tensor can be determined by updating $\theta_{\mathcal{A}}$. At the tth iteration, the $\theta_{\mathcal{A}}$ subproblem is formulated as

$$\begin{aligned} &\min_{\theta_{\mathcal{A}}} \left\|\mathcal{Y}-\mathcal{X}^{t+1}-\mathcal{A}\right\|_{\mathrm{F}}^{2}+\gamma_{2}\|\mathcal{A}\|_{\mathrm{AOS}} \\ &\text{s.t. } \mathcal{A}=f_{\theta_{\mathcal{A}}}(\mathbf{z}). \end{aligned} \tag{8}$$

Similar to the problem (6), we leverage Adam to solve the problem (8). Then, the anomaly tensor can be obtained

$$\mathcal{A}^{t+1}=f_{\theta_{\mathcal{A}}^{t+1}}(\mathbf{z}). \tag{9}$$

Once the preset iteration T is reached, the anomaly detection map $\mathbf{M}\in\mathbb{R}^{n_1\times n_2}$ is computed as

$$\mathbf{M}=\frac{\overline{\mathbf{M}}-\min\left(\overline{\mathbf{M}}\right)}{\max\left(\overline{\mathbf{M}}\right)-\min\left(\overline{\mathbf{M}}\right)} \tag{10}$$

where $\overline{\mathbf{M}}(i,j)=\|\mathcal{A}(i,j,:)\|_{\mathrm{F}}$ with $i=1,2,\ldots,n_1$ and $j=1,2,\ldots,n_2$.

The main steps of the proposed CBAR method for HAD are outlined in Algorithm 1.

Algorithm 1 Proposed CBAR Method for HAD

Input: $\mathcal{Y}$, $K_1,K_2,K_3,K_4,\gamma_1,\gamma_2,(r_1,r_2,r_3)$ and T;

1: **Initialize:** $t=0$;
2: **while** $t\le T$ **do**
3: update $\theta_{\mathcal{X}}$ via (6);
4: update $\theta_{\mathcal{A}}$ via (8);
5: $t=t+1$;
6: **end while**
7: **return M** via Eq. (10);

Output: **M**;

IV. Experiments and Discussion

In this section, we conduct a series of experiments to validate the effectiveness of the proposed CBAR model on six benchmark datasets. The comparison methods include RX [63], CRD [27], GTVLRR [34], PTA [35], LARTVAD [36], PCA-TLRSR [64], GAED [65], and Auto-AD [46]. Specifically, RX is a classical statistical-based method; CRD, GTVLRR, PTA, LARTVAD, and PCA-TLRSR are model-based methods; and Auto-AD and GAED are deep learning-based methods. Optimal parameter settings were selected for each dataset based on the detection performance of each method. In the following sections, we first introduce the experimental datasets. Then, we analyze the evaluation metrics and the experimental results from both qualitative and quantitative perspectives. Finally, we discuss the contributions of key components within the proposed CBAR model.

A. Experimental Datasets

In the experiments, six benchmark datasets are employed. The pseudo-color images and corresponding ground truth maps are shown in Fig. 5. More specific details about these datasets are provided as follows.

1) Airport: The Airport datasets were acquired using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS), where aircraft are regarded as anomalies. The Airport-4 dataset consists of 191 spectral bands with a spatial resolution of 100×100, while the Sandiego dataset has a spatial resolution of 100×100 with 189 spectral bands. The pseudo-color images and corresponding ground truth maps are shown in Fig. 5(a) and (b).

2) Urban: The Urban datasets were captured using AVIRIS, where roofs are identified as anomalies. The dimensions of Urban-3 and Urban-4 datasets are $100\times100\times191$ and $100\times100\times205$, respectively. The pseudo-color images and corresponding ground truth maps are shown in Fig. 5(c) and (d).

3) Beach: The Beach-4 dataset was captured by the ROSIS-03 sensor, with a spatial resolution of 102×150 and 150

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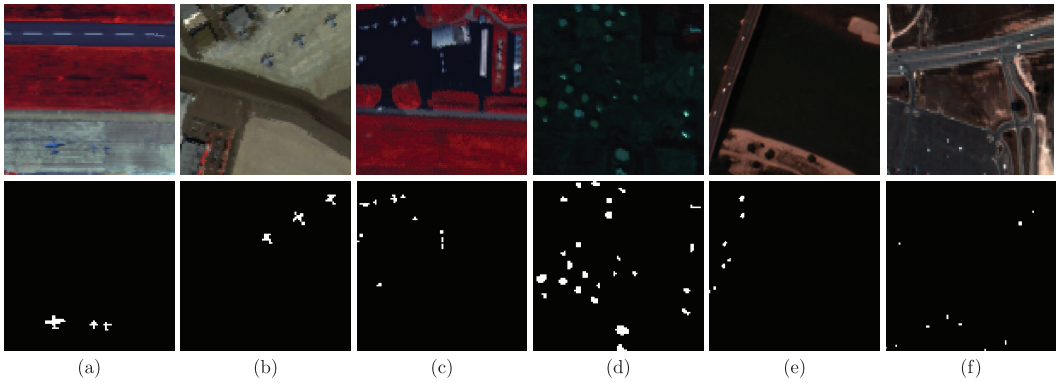


Fig. 5. Pseudo-color images and ground truth maps of the six benchmark datasets. The first row presents the pseudo-color images by selecting bands (40, 32, 20), while the second row shows the corresponding ground truth. (a) Airport-4. (b) Sandiego. (c) Urban-3. (d) Urban-4. (e) Beach-4. (f) HYDICE.

spectral bands. The cars on the bridge are treated as anomalies. The pseudo-color image and corresponding ground truth map are shown in Fig. 5(e).

4) Hyperspectral Digital Imagery Collection Experiment: The Hyperspectral Digital Imagery Collection Experiment (HYDICE) dataset, with dimensions of 100× 80 ×162, was collected using the HYDICE airborne sensor. Vehicles and roofs are identified as anomalies. The pseudo-color image and corresponding ground truth map are shown in Fig. 5(f).

B. Evaluation Metrics

To effectively evaluate the performance of HAD methods, the receiver operating characteristic (ROC) curve and the area under the curve (AUC) are commonly used. In this article, we employ ROC curve of $(\mathrm{P_d}, \mathrm{P_f})$ to qualitatively represent overall detection accuracy, where $\mathrm{P_d}$ denotes the detection probability and $\mathrm{P_f}$ represents the false alarm probability. Points on the ROC curve, defined as $\mathrm{P_d^{(\tau)}}$ and $\mathrm{P_f^{(\tau)}}$, correspond to a given threshold τ and calculated as follows:

$$\mathrm{P_d^{(\tau)}} = \frac{\mathrm{TP}^{(\tau)}}{N_\mathrm{a}}, \quad \mathrm{P_f^{(\tau)}} = \frac{\mathrm{FP}^{(\tau)}}{N_\mathrm{b}} \tag{11}$$

where $\mathrm{TP}^{(\tau)}$ is the number of anomaly pixels that are correctly identified as anomalies at threshold τ, and $\mathrm{FP}^{(\tau)}$ is the number of background pixels that are incorrectly classified as anomalies. N_a and N_b denote the total number of anomaly and background pixels in the ground truth, respectively. The corresponding AUC$(\mathrm{P_d}, \mathrm{P_f})$ value serves as a quantitative measure of detection performance. The closer the AUC$(\mathrm{P_d}, \mathrm{P_f})$ value is to 1, the better the detection performance.

Moreover, the separability boxplots illustrate the statistical distribution of anomalous response values at pixel locations for both background and anomaly categories. Two boxplots are shown: one in red representing anomalies and the other in blue representing the background, with each spanning the 10%–90% statistical range [66]. For a given method, if the background box (blue) is located at the bottom and is narrow in width, without overlapping with the anomaly box (red), it signifies strong background suppression and effective separation between the background and anomalies.

C. Detection Performance for Different Methods

This part evaluates the performance of the proposed method by comparing it with eight state-of-the-art methods across six benchmark datasets.

The anomaly detection maps of nine HAD methods across six datasets are presented in Fig. 6. By comparing with the ground truth, the anomaly detection map can directly reflect both the anomaly detection performance and degree of background suppression. As shown in Fig. 6, the anomaly detection maps obtained by the proposed CBAR model are closer to the ground truth than those produced by other methods. Specifically, the proposed model demonstrates superior accuracy in detecting anomalies while effectively suppressing background pixels. Taking the Sandiego dataset as an example, the RX method exhibits limited effectiveness, with anomalous targets barely visible in the resulting maps. Although model-based methods (including CRD, GTVLRR, PTA, LARTVAD, and PCA-TLRSR) can detect some anomalies, the detection anomaly maps often retain background pixels. Among deep learning-based methods, Auto-AD effectively suppresses background pixels in detection maps but frequently misses anomalous pixels. GAED achieves a better balance between background suppression and anomaly detection than Auto-AD, yet still falls short of producing satisfactory visual results. Similar trends are observed across other datasets, where the proposed CBAR model consistently achieves superior performance in anomaly detection and background pixel suppression.

Another example is the Urban-4 dataset, where RX, CRD, GTVLRR, Auto-AD, and GAED fail to perform well, as the target shapes are almost invisible in the anomaly detection maps. PTA and PCA-TLRSR can highlight the anomalous targets, but they also outline the background boundaries, making it difficult to distinguish between the background and anomalies. In contrast, the proposed CBAR model effectively detects the structure of anomalous targets while successfully avoiding background interference. Similar detection results are also observed in other datasets. The outstanding detection performance of CBAR can be attributed to the strong representation capability of the proposed CBAR model.

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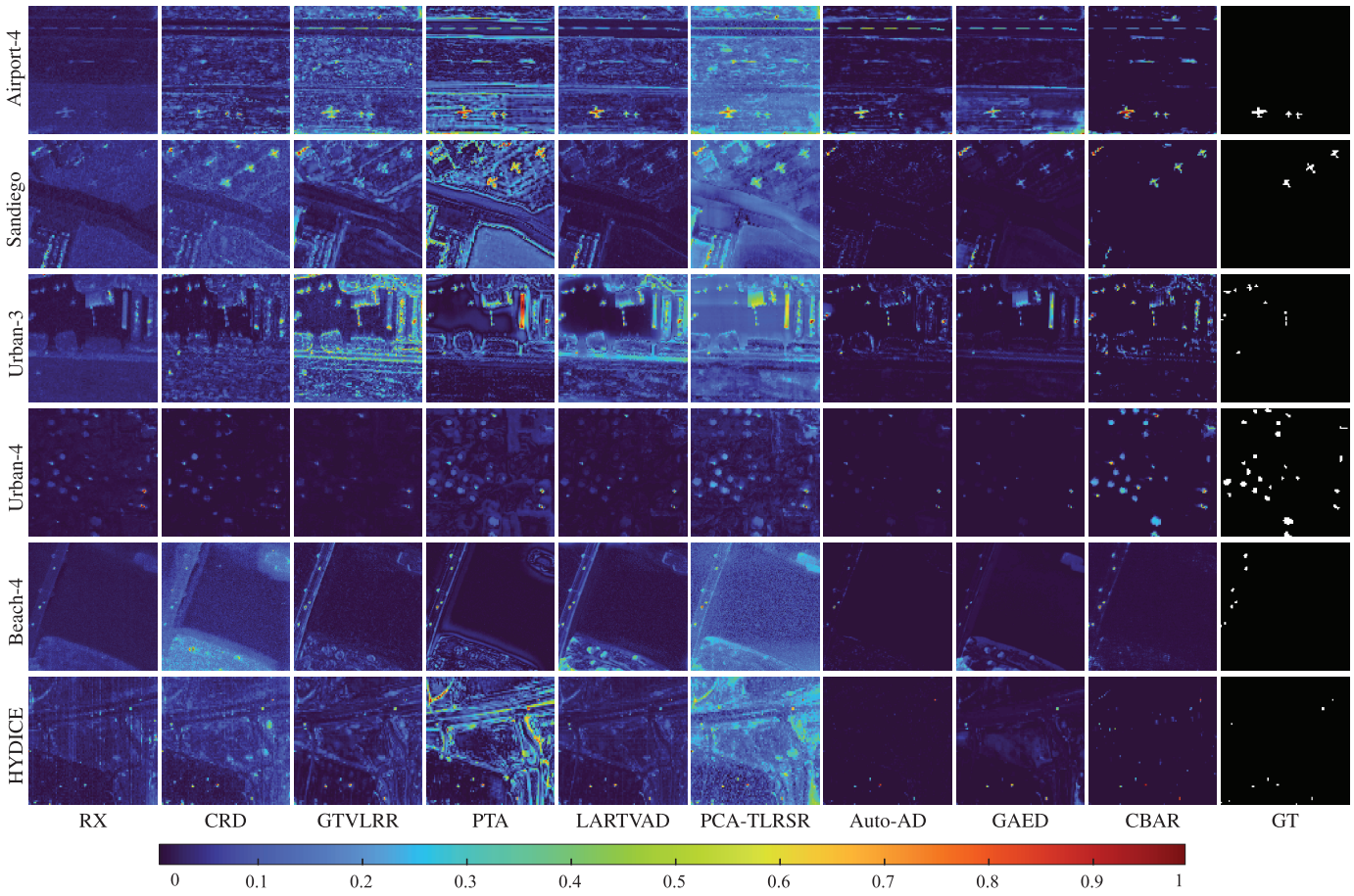


Fig. 6. Anomaly detection maps for different methods on six benchmark datasets.

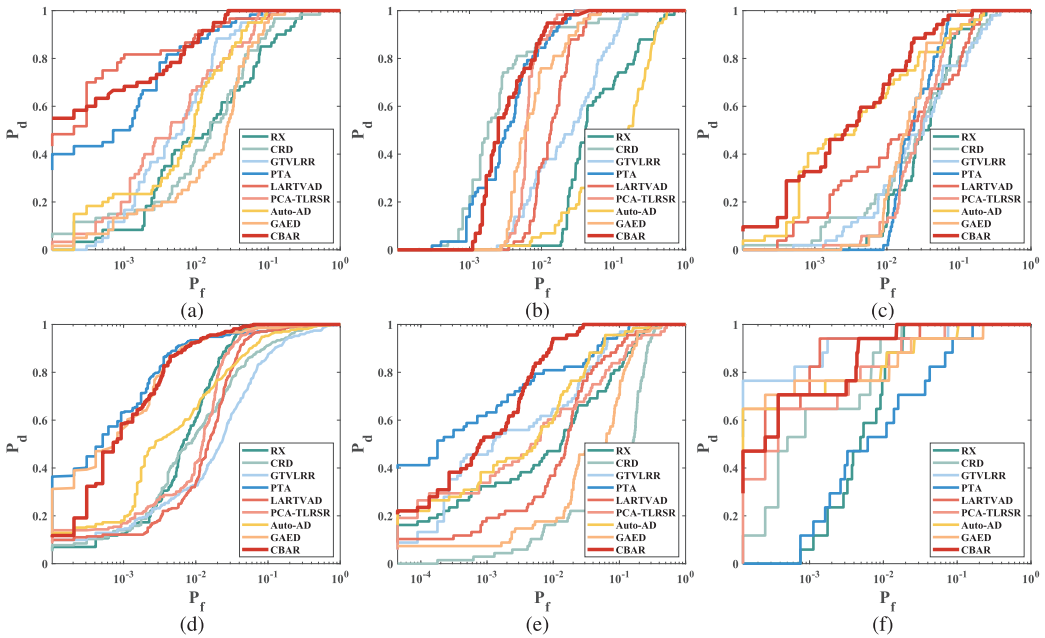


Fig. 7. ROC curves of (P_d, P_f) for different methods on six benchmark datasets. (a) Airport-4. (b) Sandiego. (c) Urban-3. (d) Urban-4. (e) Beach-4. (f) HYDICE.

To provide a more comprehensive comparison of each method, Fig. 7 presents the ROC curves of (P_d, P_f) for nine HAD methods across six datasets. As shown in Fig. 7, the curves of the proposed model are positioned above the other curves and closest to the top left corner, indicating superior overall detection performance. Based on these ROC curves, the AUC(P_d, P_f) values for each method are summarized in Table I, where the best and second-best values are highlighted

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TABLE I
AUC(P_d, P_f) VALUES AND DETECTION TIME OF NINE HAD MODELS ON DIFFERENT DATASETS

Datasets	Metric	RX	CRD	GTVLRR	PTA	LARTVAD	PCA-TLRSR	Auto-AD	GAED	CBAR
Airport-4	AUC	0.9526	0.9599	0.9875	0.9945	0.9940	0.9850	0.9838	0.9679	**0.9959**
	Time(s)	**0.01**	2.38	73.98	14.13	30.12	5.82	4.91	22.69	14.71
Sandiego	AUC	0.8885	0.9893	0.9601	0.9878	0.9831	0.9927	0.8234	0.9889	**0.9950**
	Time(s)	**0.04**	4.23	70.31	11.65	19.41	4.32	21.85	21.85	17.84
Urban-3	AUC	0.9513	0.9393	0.9348	0.9690	0.9506	0.9605	0.9755	0.9752	**0.9871**
	Time(s)	**0.18**	5.41	98.25	14.61	27.22	1.22	12.55	22.79	32.04
Urban-4	AUC	0.9887	0.9655	0.9433	0.9912	0.9759	0.9851	0.9747	0.9924	**0.9966**
	Time(s)	**0.05**	5.87	54.41	12.95	18.62	2.15	22.26	35.13	11.59
Beach-4	AUC	0.9538	0.8571	0.9806	0.9868	0.9672	0.9516	0.9803	0.9286	**0.9961**
	Time(s)	**0.04**	10.76	226.05	15.14	23.31	4.46	21.71	24.22	13.19
HYDICE	AUC	0.9932	0.9967	0.9953	0.9730	0.9978	0.9935	0.9910	0.9837	**0.9981**
	Time(s)	**0.07**	0.97	123.29	14.13	30.09	5.82	21.72	22.69	15.55
Average	AUC	0.9547	0.9513	0.9669	0.9837	0.9781	0.9781	0.9548	0.9728	**0.9940**
	Time(s)	**0.07**	4.94	107.72	13.77	24.80	3.97	17.50	24.90	17.49

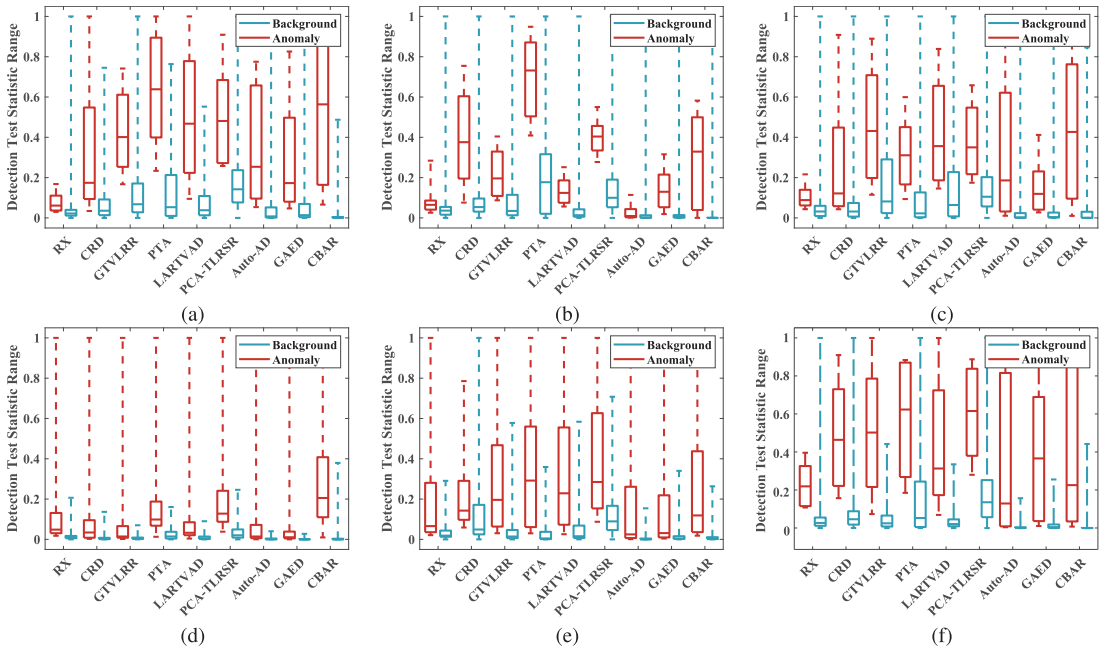


Fig. 8. Separability boxplots for different methods on six benchmark datasets. (a) Airport-4. (b) Sandiego. (c) Urban-3. (d) Urban-4. (e) Beach-4. (f) HYDICE.

in bold and underlined, respectively. As shown in Table I, the proposed CBAR method achieves the highest AUC(P_d, P_f) values and average AUC(P_d, P_f) values among all other methods across six benchmark datasets, demonstrating superior anomaly detection accuracy. More metric indices related to ROC and AUC can be found in the supplementary material.

To further evaluate the effectiveness of distinguishing between the background and anomalies, as well as its capability in suppressing background pixels, Fig. 8 presents the separability boxplots for different methods across six benchmark datasets. As shown in Fig. 8, the proposed CBAR model exhibits the narrowest background box across all datasets compared to other methods, with a mean value near zero, particularly for the Airport-4, Urban-4, and HYDICE datasets. This indicates that the proposed model has the best background suppression capability. Moreover, the proposed CBAR model demonstrates higher mean values for the anomaly boxplots and no overlap between the anomaly and background boxes across all datasets compared to other methods, indicating its superior ability to distinguish anomalies from the background.

By comparing the detection times of different methods across six benchmark datasets, it is observed that the RX method achieves the fastest detection speed. Although the proposed CBAR model requires more time than RX, CRD, and PCA-TLRSR, it remains more efficient than GTVLRR, LARTVAD, and GAED. Table II summarizes the number of learnable parameters for two deep learning-based methods and the proposed CBAR model on the Airport-4 dataset.

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TABLE II
PARAMETERS OF TWO DEEP LEARNING-BASED METHODS AND THE PROPOSED CBAR MODEL ON THE AIRPORT-4 DATASET

Params (M)	Auto-AD	GAED	CBAR
Background	3.19	0.05	0.04
Anomalies*	–	–	32.48

* Auto-AD and GAED do not consider the anomaly representation.

While CBAR incorporates deep representation of anomalies and thus contains more parameters than Auto-AD and GAED, its overall model complexity remains within an acceptable range.

D. Ablation Study

The key components of the proposed CBAR model include the continuous background representation, the deep anomaly representation, and two terms of overfit shield (i.e., BOS and AOS). To assess the contribution of each component, we conduct comprehensive experiments.

1) Contribution of the Continuous Background Representation: To evaluate the contribution of the continuous background representation, we replace the low-rank tensor function in the CBAR model with a learnable parameterized tensor, referred to as CBAR without CBR. As shown in Table III, the CBAR without CBR model exhibits lower detection accuracy compared to the proposed CBAR model, as indicated by lower AUC(P_d, P_f) values. The superior performance of the proposed CBAR model can be attributed to the continuous background representation, which characterizes the background from a continuous perspective and effectively captures both the low rankness and smoothness of the background.

2) Contribution of the Deep Anomaly Representation: To evaluate the contribution of the deep anomaly representation, we replace it with a learnable parameterized tensor in the CBAR model, referred to as CBAR without DAR. As shown in Table III, the CBAR without DAR performs competitively on small-scale anomalous targets, such as Urban-3, Beach-4, and HYDICE, but exhibits lower AUC(P_d, P_f) values on datasets with large-scale anomalies such as Airport-4, indicating inferior detection accuracy compared to the proposed CBAR model. The deep anomaly representation excels at capturing the complex geometric structure of anomalies, thereby enhances anomaly detection accuracy.

3) Contribution of BOS: To evaluate the contribution of BOS, we compare the CBAR with and without BOS. As shown in Table III, the CBAR without BOS (CBAR w/o BOS) leads to a notable reduction in AUC(P_d, P_f) values across all datasets, particularly those with large-scale anomalies. Furthermore, we present the history of AUC(P_d, P_f) and the corresponding pseudo-color background images during iterations for CBAR and CBAR w/o BOS on Airport-4 in Fig. 9. From this figure, it can be observed that the CBAR w/o BOS incorporates anomaly pixels (e.g., airplanes) into the background during the iterations. The CBAR model effectively mitigates this overfitting. Additionally, the CBAR model converges faster and achieves a higher AUC(P_d, P_f) value compared to the CBAR w/o BOS model.

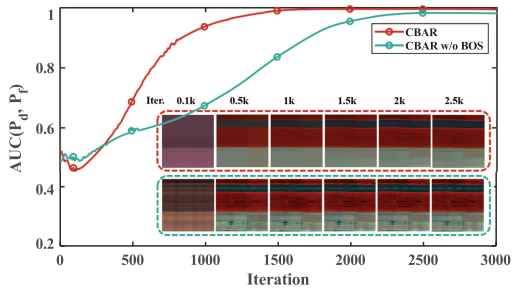


Fig. 9. History of AUC(P_d, P_f) values and corresponding pseudo-color images of the background of CBAR and CBAR without BOS on Airport-4.

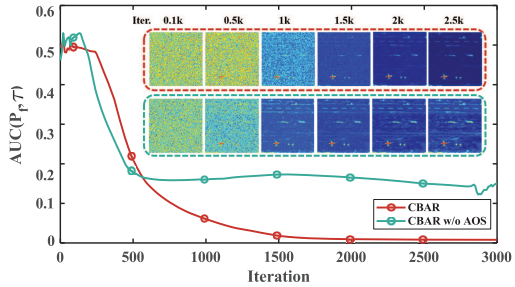


Fig. 10. History of AUC(P_f, τ) values and corresponding anomaly detection maps of CBAR and CBAR without AOS on Airport-4.

4) Contribution of AOS: To evaluate the contribution of AOS, we compare the proposed CBAR model with and without the AOS. As shown in Table III, the CBAR without AOS (CBAR w/o AOS) exhibits in lower AUC(P_d, P_f) values than the CBAR model across all datasets except Airport-4, indicating a general decline in overall detection performance. Furthermore, we utilize the AUC(P_f, τ) value [69] to quantitatively measure the degree of background pixel suppression, where a value closer to 0 indicates stronger suppression of background pixels. Fig. 10 illustrates the history of AUC(P_f, τ) values and corresponding anomaly detection maps during the iterations of CBAR and CBAR w/o AOS on Airport-4, and it is evident that the CBAR w/o AOS incorporates background pixels (e.g., lane dividing lines) into anomaly detection maps. Conversely, the CBAR model effectively mitigates this overfitting.

E. Discussion

In this section, we analyze the influence of different background and anomaly representations on anomaly detection performance.

1) Influence of Different Low-Rank Background Representations: To analyze the influence of the continuous background representation, we compare it with the discrete Tucker and discrete CP decompositions to characterize the background. The AUC(P_d, P_f) values on six datasets are shown in Table IV. From this table, we observe that the background representation method based on the continuous

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TABLE III

AUC(P_d, P_f) VALUES OF THE CBAR MODEL AND ITS VARIANTS ACROSS DIFFERENT DATASETS

Models	Airport-4	Urban-3	Beach-4	HYDICE	Average
CBAR w/o CBR	0.9901	0.9745	0.9835	0.9943	0.9856
CBAR w/o DAR	0.9879	0.9808	0.9951	0.9978	0.9904
CBAR w/o BOS	0.9838	0.9713	0.9955	0.9914	0.9855
CBAR w/o AOS	**0.9962**	0.9668	0.9739	0.9878	0.9812
CBAR	0.9959	**0.9871**	**0.9961**	**0.9981**	**0.9943**

TABLE IV

AUC(P_d, P_f) VALUES OF DIFFERENT LOW-RANK REPRESENTATIONS OF BACKGROUND ACROSS DIFFERENT DATASETS

Background Representation	Airport-4	Sandiego	Urabn-3	Urabn-4	Beach-4	HYDICE
Discrete CP Decomposition [67]	0.9951	0.9947	0.9759	0.9947	0.9943	0.9958
Discrete Tucker Decomposition [68]	0.9955	0.9916	0.9310	0.9941	0.9920	0.9812
Low-rank Tensor Function	**0.9959**	**0.9950**	**0.9871**	**0.9966**	**0.9961**	**0.9981**

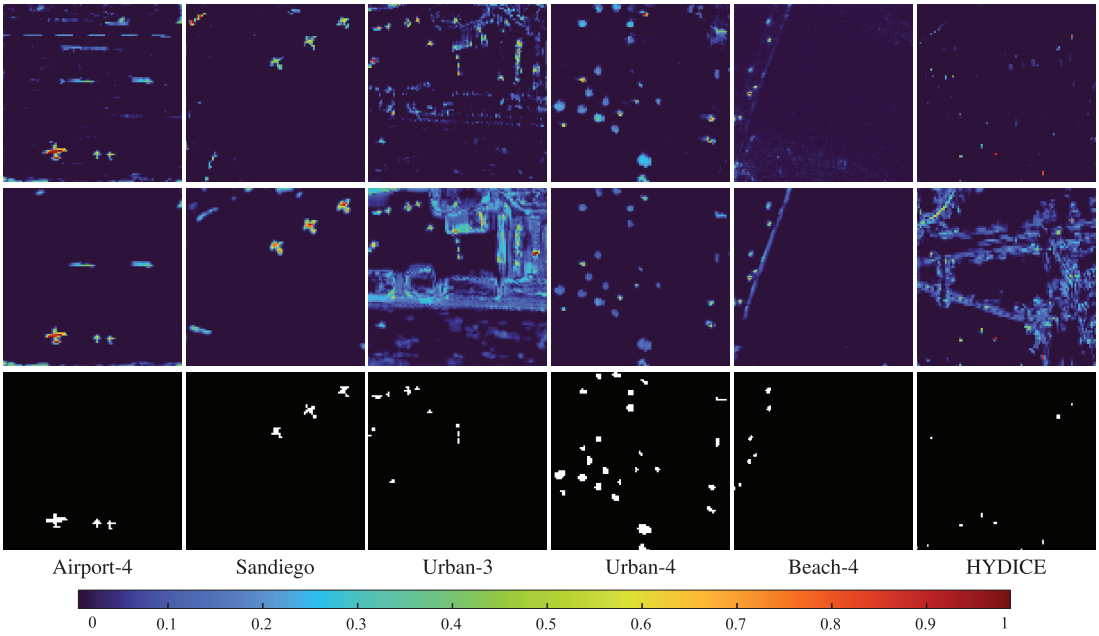


Fig. 11. Anomaly detection maps on six benchmark datasets. (Top to bottom) Anomaly detection maps obtained by MLP-based CBAR, CNN-based CBAR, and ground truth.

background representation exhibits a clear advantage over both discrete Tucker and discrete CP decompositions in terms of AUC(P_d, P_f) values. This can be attributed to the continuous characterization of the background, which enables the proposed model to leverage both its low-rankness and smoothness simultaneously.

2) Influence of Different Network Architectures in Anomaly Representation: To analyze the influence of different network architectures in anomaly representation, we consider two popular architectures, i.e., MLP and CNN. Specifically, we leverage MLP and CNN to separately capture anomalies, referred to as MLP-based and CNN-based CBAR. The deep anomaly representation in CNN-based CBAR is defined as

$$\mathcal{A} = f_{\theta_{\mathcal{A}}}(\mathcal{Z}) = \mathrm{Conv}_l(\sigma(\mathrm{Conv}_{l-1}(\cdots\sigma(\mathrm{Conv}_1(\mathcal{Z})))))$$

where l denotes the number of convolutional layers, $\mathrm{Conv}_i (i = 1, 2, \ldots, l)$ is the convolution operation, $\sigma(\cdot)$ is the ReLU activation function, and $\theta_{\mathcal{A}}$ is the network parameter. The deep anomaly representation in CNN-based CBAR $f_{\theta_{\mathcal{A}}}(\cdot)$ maps a random tensor $\mathcal{Z} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ to the anomaly tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$.

As shown in Fig. 11, the anomaly detection maps detected by the CNN-based CBAR model more closely match the ground truth for large-scale anomalies compared to the MLP-based CBAR model. However, for smaller anomalies, the CNN-based CBAR model struggles to effectively separate anomaly pixels from the background. As shown in Table V, the MLP-based CBAR model exhibits significantly higher performance on datasets with small-scale anomalies, such as Urban-3 and HYDICE, while achieving comparable AUC (P_d, P_f) values on datasets with large-scale anomalies, including Airport-4 and Sandiego. Therefore, for simplicity, we adopt MLP as the backbone architecture of deep anomaly representation in our work.

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TABLE V
AUC(P_d, P_f) VALUES OF DIFFERENT NETWORK ARCHITECTURES FOR ANOMALY DETECTION ACROSS DIFFERENT DATASETS

Network Architecture	Airport-4	Sandiego	Urabn-3	Urabn-4	Beach-4	HYDICE
MLP	0.9959	0.9949	**0.9871**	0.9966	0.9961	**0.9981**
CNN	**0.9975**	**0.9973**	0.9686	**0.9973**	**0.9963**	0.9954

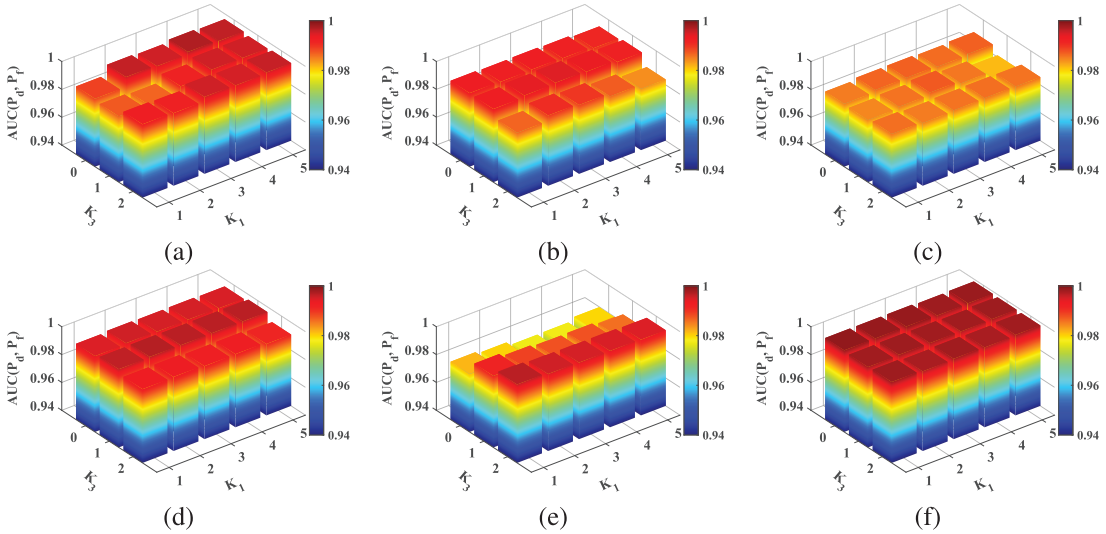


Fig. 12. AUC(P_d, P_f) values of the proposed CBAR model under different K_1 and K_3 settings across six benchmark datasets. (a) Airport-4. (b) Sandiego. (c) Urban-3. (d) Urban-4. (e) Beach-4. (f) HYDICE.

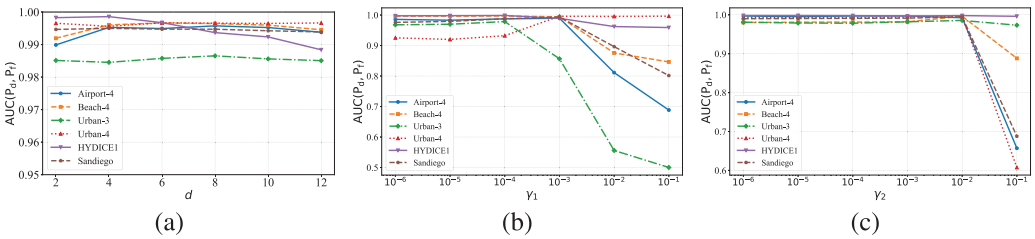


Fig. 13. AUC(P_d, P_f) values of the proposed CBAR model with different parameter settings across six benchmark datasets. (a) Scaling factor d. (b) Tradeoff parameter γ_1. (c) Tradeoff parameter γ_2.

F. Hyperparameter Analysis

The proposed CBAR model involves three types of hyperparameter: 1) the group sizes in BOS and AOS, i.e., K_1, K_2, K_3, and K_4; 2) the background rank, i.e., (r_1, r_2, r_3); and 3) the tradeoff parameters, i.e., γ_1 and γ_2.

For the group sizes, we set $K_1 = K_2$ and $K_3 = K_4$ for simplicity. As shown in Fig. 12, it can be observed that K_1 and K_3 significantly influence the detection results. For different datasets, we can have an optimal detection performance with moderate group sizes.

For the background rank (r_1, r_2, r_3), we set $r_1 = \lfloor n_1/d \rfloor$, $r_2 = \lfloor n_2/d \rfloor$, and $r_3 = \lfloor n_3/d \rfloor$, where d is a scaling factor and $\lfloor \cdot \rfloor$ denotes the floor function. Therefore, we analyze the effect of the scaling factor d on detection performance. As shown in Fig. 13(a), it can be observed that the impact of the scaling factor d varies across different datasets. Nevertheless, the CBAR model generally exhibits relative robustness with respect to the scaling factor d across most datasets.

The parameters γ_1 and γ_2 are tradeoff weights for BOS and AOS, respectively. As shown in Fig. 13(b), the detection performance improves when γ_1 ranges from 10^{-6} to 10^{-4}. However, when γ_1 becomes too large, the detection accuracy begins to decline, particularly beyond 10^{-3}, as an excessively strong BOS term may hinder the reconstruction of the background. Similarly, as depicted in Fig. 13(c), the CBAR model remains relatively robust when γ_2 is less than 10^{-2}. Nevertheless, as γ_2 increases beyond a reasonable level, the AOS term may suppress valid anomaly representations, thereby leading to a decline in detection performance.

V. CONCLUSION

In this article, we have proposed an unsupervised HAD model that organically integrates the continuous background representation and the deep anomaly representation. For the background, we first characterized the background from a continuous perspective, which simultaneously captures both low rankness and smoothness of the background. For anomalies, we leveraged the deep anomaly representation to capture the complex geometric structure of anomalies. Additionally, we designed two terms of overfit-shield, i.e., BOS and

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AOS, which mitigate the overfitting of the background and anomalies to the observed HSI, thereby further ensuring the detection performance. To solve the CBAR model, we developed an alternating minimization-based algorithm. Extensive experiments demonstrate the effectiveness of the proposed method.

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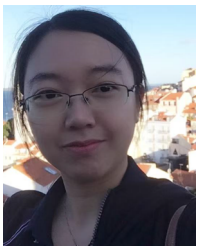
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